

POLITECNICO DI TORINO



**Politecnico
di Torino**

Lab 4

Music and multimedia streaming over the Internet

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1 Introduction

The aim of this lab is to familiarize students with fundamental Internet concepts. Specifically, the goal is to acquire expertise in network, transport, and HTTP protocols, as well as to explore basic tools for testing, analyzing, and evaluating network behavior. .

2 Preliminary operations

Access your Crownlabs account. Open the **Lab 2** virtual machine, then, inside it, open the Wireshark tool, and the terminal emulator, which are the required tools for this lab activity. In Wireshark, put the following string in the capture filters text box:

```
not tcp port 5901 and not tcp port 6080 and not tcp port 80 and not  
tcp port 443
```

This is necessary to prevent capturing traffic related to VM control and web-related operations. While some background noise may still be present, it will be minimal. You can then begin capturing on the "Loopback" and "enp1s0" interfaces, which will record traffic sent by the system to itself and through the virtual Ethernet connection named "enp1s0".

3 Delivery

You will receive both a LaTeX template zip and a PDF file containing a set of questions to answer. You can choose to directly modify the LaTeX document (recommended) by using tools like Overleaf, or create a separate document starting from the PDF and reporting both the original questions and the answers. Feel free to include screenshots wherever they may be helpful.

4 Exercises

4.1 Ping

4.1.1 To known IP address

Choose an IP from the following list at random:

- 130.192.9.250
- 130.192.181.193
- 130.192.9.17

Execute `ping`, sending 3 packets, using one single command, to the chosen IP.

In order to send exactly 3 packets, you can read the manual of the `ping` command, available typing `man ping` in the terminal or by searching the manual of ping in the Web.

Using Wireshark, identify the packet types involved in the process.

4.1.2 To named host

Repeat the previous exercise, but replace the IP address with one of the following server names:

- leonardo.polito.it

- `giove.polito.it`
- `ns3.polito.it`

4.2 Netcat

4.2.1 TCP

In a terminal, run Netcat (`nc` or `netcat` command equivalently) with the following setup:

- **listen** mode (makes it a server accepting connections)
- **TCP** protocol (default) on IPv4 localhost **only**
- **host**: use `localhost` or `127.0.0.1` as host to bind to.
- **port** number: `8080` .

Check the manual page with its examples to build the correct command.

In a separate terminal, run the Netcat command as a client to connect to the previously started server. Exchange simple sentences between the two terminals, then close the client first. Use Wireshark to monitor the connection initiation and termination processes, as well as the packets containing the exchanged messages.

4.2.2 UDP

Replicate the example of sec. 4.2.1 substituting the TCP protocol with UDP, sending the first messages in the direction server (listener) → client.

4.3 Web

4.3.1 HTTP Server

The `test_server.py` script utilizes Python's built-in web server and is modified to serve files from the `$HOME/test_server` directory. It also handles **POST** requests sent to the `/login` path, expecting a JSON object with two fields: `user`, (representing the username) and `pwd` (containing the password). The folder contains only an `index.html` file with a very simple login page, which provides a simple login page that submits a **POST** request when the login button is pressed.

1. Run Wireshark.
2. Run the server in a terminal.
3. Open the page (`http://localhost:8080`) with the browser in the VM and try to enter some credentials.
4. Press the login button and check the terminal to verify that the server correctly received the username and password.
5. Stop the server.
6. Close the browser.
7. Use Wireshark to observe the network traffic generated during the login process. In the packet payloads, look for the username and password you entered in the browser.

4.3.2 HTTPs Server

Repeat the same operations of sec. [4.3.1](#). Replace the `test_server.py` with the `test_server_https.py` (remember to use https instead of http in the browser and to choose "accept the risk and continue" when opening the page).